

PHYSICS

Topic: Newton Laws of Motion | Grade: Grade 10

Total Marks: **30**

Time Allowed: **25 minutes**

STUDENT INFORMATION

Name

Roll Number

Section

SECTION A (9 MARKS)

Choose the correct answer. Each question carries 3 marks.

Q1. According to Newton's Second Law of Motion, which of the following combinations will produce the greatest acceleration?

A) Force = 5 N, Mass = 2 kg

B) Force = 10 N, Mass = 5 kg

C) Force = 10 N, Mass = 2 kg

D) Force = 15 N, Mass = 3 kg

Easy [3 marks]

Q2. Which of the following is the best example of inertia?

A) A stationary ball starting to roll when kicked

B) A passenger in a bus lurching forward when the bus suddenly stops

C) A rocket launching upwards into space

D) A magnet attracting iron filings

Easy [3 marks]

Q3. A boy pushes a wall with a force of 50 N. According to Newton's Third Law, the wall pushes back on the boy with a force of:

A) 0 N

B) 25 N

C) 50 N

D) 100 N

Easy [3 marks]

SECTION B (12 MARKS)

Answer the following questions. Show all working.

Q1. A force of 20 N acts on a body of mass 5 kg. Calculate: (a) The acceleration produced (b) The velocity of the body after 3 seconds if it starts from rest.

Medium [6 marks]

Q2. State and explain Newton's Third Law of Motion. Provide two real-life examples to illustrate this law.

Medium [6 marks]

SECTION C (9 MARKS)

Answer the following question with complete working and reasoning.

Q1. A wooden block of mass 10 kg rests on a horizontal surface. The coefficient of static friction between the block and surface is 0.3, and the coefficient of kinetic friction is 0.2. A horizontal force of 50 N is applied to the block. (a) Calculate the maximum static friction force. Will the block start moving? Support your answer with calculations. (b) If the block is already in motion, find its acceleration. (Take $g = 10 \text{ m/s}^2$)

Hard

[9 marks]